

# HOMework 6

## ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03), SPRING, 2018

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### SECTION 4.1 - VECTOR SPACES AND SUBSPACES

1. Let  $V$  be the set of all polynomials in  $\mathbb{P}_n$  such that  $p(0) = 0$ , i.e.

$$V = \{p \in \mathbb{P}_n : p(0) = 0\}.$$

Show that  $V$  is a subspace of  $\mathbb{P}_n$ .

Recall that a subspace of a vector space  $W$  is a subset  $H$  of  $W$  such that:

- (a)  $0 \in H$ ,
- (b)  $H$  is closed under addition,
- (c)  $H$  is closed under scalar multiplication.

So, to show that  $V$  is a subspace of  $\mathbb{P}_n$ , we need only show that  $V$  satisfies the three requirements above.

The first requirement is easy. Since the zero polynomial is zero when evaluated at any point, it is certainly zero when evaluated at 0. So,  $0 \in V$ .

To see that the second condition holds, note that if  $p_1, p_2 \in V$ , then

$$(p_1 + p_2)(0) = p_1(0) + p_2(0) = 0 + 0 = 0.$$

So, the sum of two elements of  $V$  is in  $V$  and we may conclude that  $V$  is closed under vector addition.

For the third condition, observe that if  $p \in V$  and  $k \in \mathbb{R}$ , then

$$(kp)(0) = k(p(0)) = k(0) = 0.$$

So, a scalar multiple of an element of  $V$  is in  $V$  and we may conclude  $V$  is also closed under scalar multiplication. Therefore,  $V$  is a subspace of  $\mathbb{P}_n$ .

2. The set of all continuous real-valued functions defined on a closed interval  $[a, b]$  in  $\mathbb{R}$  is denoted by  $C[a, b]$ . This set is a subspace of the vector space of all real-valued functions defined on  $[a, b]$  (this vector space is infinite-dimensional, but don't let that concern you). Let  $V = \{f \in C[a, b] : f(a) = f(b)\}$ . Show that  $V$  is a subspace of  $C[a, b]$ .

To show that  $V = \{f \in C[a, b] : f(a) = f(b)\}$  is a subspace of the vector space of all real-valued functions defined on  $[a, b]$ , we need only show that  $V$  satisfies the three requirements listed in problem 1.

For the first requirement, observe that the zero function on the interval  $[a, b]$  (that is, the function which is zero at every value of  $x$  in the interval  $[a, b]$ ) is in  $V$ , as the zero function is continuous and certainly is zero at both  $a$  and  $b$ .

For the second requirement, observe that the statement  $f(a) = f(b)$  is equivalent to the statement  $f(a) - f(b) = 0$ . So, if  $f_1$  and  $f_2$  are elements of  $V$ , then  $f_1(a) - f_1(b) = 0$  and  $f_2(a) - f_2(b) = 0$ , and we have

$$\begin{aligned}(f_1 + f_2)(a) - (f_1 + f_2)(b) &= (f_1(a) + f_2(a)) - (f_1(b) + f_2(b)) \\&= f_1(a) + f_2(a) - f_1(b) - f_2(b) \\&= (f_1(a) - f_1(b)) + (f_2(a) - f_2(b)) \\&= 0 + 0 = 0.\end{aligned}$$

So,  $f_1 + f_2 \in V$  and we may conclude that  $V$  is closed under scalar multiplication.

For the third and final requirement, observe that if  $f \in V$  and  $c \in \mathbb{R}$ , then

$$(cf)(a) - (cf)(b) = cf(a) - cf(b) = c(f(a) - f(b)) = c(0) = 0.$$

So,  $cf \in V$  as well and we may conclude  $V$  is closed under scalar multiplication. Therefore,  $V$  is a subspace of  $C[a, b]$ .

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## SECTION 4.2 - NULL SPACES, COLUMN SPACES, AND LINEAR TRANSFORMATIONS

3. Let  $M_{2 \times 2}(\mathbb{R})$  be the vector space of all  $2 \times 2$  matrices over  $\mathbb{R}$ , and define  $T : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$  by  $T(A) = A + A^T$ . Show that  $T$  is a linear transformation and give a description of the kernel of  $T$ .

To show that  $T$  is a linear transformation, we need only show that  $T(A + B) = T(A) + T(B)$  and  $T(cA) = cT(A)$  for all  $A, B \in M_{2 \times 2}(\mathbb{R})$  and all  $c \in \mathbb{R}$ . We may show that  $T(A + B) = T(A) + T(B)$  with the following calculation:

$$\begin{aligned} T(A + B) &= (A + B) + (A + B)^T \\ &= A + B + A^T + B^T \\ &= (A + A^T) + (B + B^T) \\ &= T(A) + T(B). \end{aligned}$$

Similarly, we may show  $T(cA) = cT(A)$  by noting

$$T(cA) = (cA) + (cA)^T = c(A + A^T) = cT(A).$$

So,  $T$  is a linear transformation.

To find the kernel of  $T$ , recall that

$$\ker(T) = \{A \in M_{2 \times 2}(\mathbb{R}) : T(A) = 0\}.$$

So, we need only find the elements of  $M_{2 \times 2}(\mathbb{R})$  which  $T$  maps to the zero matrix. To do so, observe that if  $T(A) = 0$ , then  $A + A^T = 0$ . Rearranging, this may be stated as  $A = -A^T$ . A matrix which satisfies this property is known as a *skew-symmetric matrix*, and may be characterized in the  $2 \times 2$  case by the following observation:

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= -\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T \\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= \begin{bmatrix} -a & -c \\ -b & -d \end{bmatrix}. \end{aligned}$$

So, since the only real number  $x$  which satisfies  $x = -x$  is  $x = 0$ , we may see that  $a = d = 0$ . Similarly, we may see from the above that we must have both  $c = -b$  and  $b = -c$ . So, we may conclude that

$$\ker(T) = \left\{ \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} : b \in \mathbb{R} \right\}.$$

4. Define  $T : C[0, 1] \rightarrow C[0, 1]$  as follows: For  $f \in C[0, 1]$ , let  $T(f)$  be the antiderivative,  $F$ , of  $f$  such that  $F(0) = 0$ . Show that  $T$  is a linear transformation and give a description of the kernel of  $T$ .

By the fundamental theorem of calculus, if  $f \in C[0, 1]$ , then  $\int_0^x f(x)dx = F(x) - F(0)$  for any antiderivative  $F$  of  $f$ . Since  $T$  sends  $f$  to the antiderivative which satisfies  $F(0) = 0$ , we then have that  $T$  is given by

$$T(f(x)) = \int_0^x f dx.$$

So, by the linearity of the integral,  $T$  is a linear transformation. To find the kernel of  $T$ , recall that

$$\ker(T) = \{f \in C[0, 1] : T(f) = 0\}.$$

So, we need only find the functions  $f \in C[0, 1]$  such that  $T(f) = 0$ . That is, we need only find the functions  $f \in C[0, 1]$  which satisfy  $\int_0^x f(x)dx = 0$  for all values of  $x \in [0, 1]$ . But, since the integral  $\int_0^x f(x)dx$  measures the net area between the graph of  $f$  and the  $x$ -axis on  $[0, x]$ , this conditions says that this area is zero for any choice of  $x \in [0, 1]$ . Thus,  $f$  must be the zero function. So, we may conclude that

$$\ker(T) = \{0\}.$$

As a side note, while you don't have to prove that  $\int_0^x f(x)dx = 0$  for all  $x \in [0, 1]$  implies  $f(x) = 0$ , here's a quick proof: Recall that if  $f(a) \neq 0$  for some value of  $a \in [0, 1]$ , then either  $f(a) > 0$  or  $f(a) < 0$  and the continuity of  $f$  yields that either  $f(x) > 0$  or  $f(x) < 0$  on the interval  $(a - \epsilon, a + \epsilon) \subseteq [0, 1]$  for some  $\epsilon > 0$ . Hence,  $\int_{a-\epsilon}^{a+\epsilon} f(x)dx \neq 0$ . So, if  $\int_0^{a+\epsilon} f(x)dx = 0$ , then

$$\begin{aligned} 0 &= \int_0^{a+\epsilon} f(x)dx \\ 0 &= \int_0^{a-\epsilon} f(x)dx + \int_{a-\epsilon}^{a+\epsilon} f(x)dx \\ \int_0^{a-\epsilon} f(x)dx &= - \int_{a-\epsilon}^{a+\epsilon} f(x)dx \neq 0. \end{aligned}$$

That is, if  $f(a) \neq 0$  for some value  $a \in [0, 1]$ , then either  $\int_0^{a+\epsilon} f(x)dx \neq 0$  or  $\int_0^{a-\epsilon} f(x)dx \neq 0$ . So, if  $f$  is continuous on  $[0, 1]$ , then  $\int_0^x f(x)dx = 0$  for all values of  $x \in [0, 1]$  if and only if  $f(x) = 0$ .

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## SECTION 4.3 - LINEARLY INDEPENDENT SETS, BASES

5. Consider the polynomials  $p_1(t) = 1 + t^2$  and  $p_2(t) = 1 - t^2$ . Is  $\{p_1, p_2\}$  a linearly independent set in  $\mathbb{P}_3$ ? Why or why not?

To determine whether  $\{p_1, p_2\}$  is a linearly independent set in  $\mathbb{P}_3$ , it will be convenient to convert  $p_1$  and  $p_2$  to coordinate vectors in the standard basis  $\mathcal{B} = \{1, t, t^2, t^3\}$  of  $\mathbb{P}_3$ . Doing so, we yield

$$[p_1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad [p_2]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Since  $[p_1]_{\mathcal{B}}$  is not a scalar multiple of  $[p_2]_{\mathcal{B}}$ , it follows that  $\{[p_1]_{\mathcal{B}}, [p_2]_{\mathcal{B}}\}$  is a linearly independent set in  $\mathbb{R}^4$ . As a set of vectors  $\{v_1, \dots, v_p\}$  belonging to a vector space  $V$  is linearly independent if and only if  $\{[v_1]_{\mathcal{B}}, \dots, [v_p]_{\mathcal{B}}\}$  is linearly independent for any basis  $\mathcal{B}$  of  $V$ , it follows that  $\{p_1, p_2\}$  is a linearly independent set in  $\mathbb{P}_3$ .

6. In the vector space of all real-valued functions, find a basis for the subspace spanned by  $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$ . (HINT: Look up the double-angle identities.)

We seek a basis for the subspace  $\text{span}\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$  of the vector space of all real valued functions. Using the double angle identities, observe that  $\sin(2t) = 2\sin(t)\cos(t)$ . Then, by the definition of span, we have

$$\begin{aligned}\text{span}\{\sin(t), \sin(2t), \sin(t)\cos(t)\} &= \{c_1 \sin(t) + c_2 \sin(2t) + c_3 \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\} \\ &= \{c_1 \sin(t) + 2c_2 \sin(t)\cos(t) + c_3 \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\} \\ &= \{c_1 \sin(t) + (2c_2 + c_3) \sin(t)\cos(t) : c_1, c_2, c_3 \in \mathbb{R}\} \\ &= \text{span}\{\sin(t), \sin(t)\cos(t)\}\end{aligned}$$

As  $\{\sin(t), \sin(t)\cos(t)\}$  spans the same space as our original set  $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$  does, it is a basis for this space if  $\{\sin(t), \sin(t)\cos(t)\}$  is linearly independent. Since  $\sin(t)$  is not a scalar multiple of  $\sin(t)\cos(t)$ , we may conclude that  $\{\sin(t), \sin(t)\cos(t)\}$  is linearly independent.

Therefore, we may conclude that  $\{\sin(t), \sin(t)\cos(t)\}$  is a basis for the subspace spanned by  $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$ .

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## SECTION 4.4 - COORDINATE SYSTEMS

7. The set  $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$  is a basis for  $\mathbb{R}^2$ . Find the change-of-coordinates matrix from the basis  $\mathcal{B}$  to the standard basis of  $\mathbb{R}^n$ .

Recall that the change of coordinates matrix from a basis  $\mathcal{B}$  of  $\mathbb{R}^n$  to the standard basis in  $\mathbb{R}^n$  is the  $n \times n$  matrix  $P_{\mathcal{B}}$  such that  $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ . So, given a vector  $x \in \mathbb{R}^2$ , the  $\mathcal{B}$ -coordinate vector of  $x$  is the vector  $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$  such that  $x = c_1 b_1 + c_2 b_2$ . Rearranging, this says  $x = \begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ . Therefore, we may conclude that the change of coordinates matrix from the basis  $\mathcal{B}$  to the standard basis of  $\mathbb{R}^n$  is given by

$$P_{\mathcal{B}} = \begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -9 & 8 \end{bmatrix}.$$

8. Let  $p_1(t) = 1 + t^2$ ,  $p_2(t) = t - 3t^2$ , and  $p_3(t) = 1 + t - 3t^2$ . The set  $\mathcal{B} = \{p_1, p_2, p_3\}$  forms a basis for

$\mathbb{P}_2$ . Find the polynomial  $q \in \mathbb{P}_2$  given that  $[q]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$ .

To find the polynomial  $q \in \mathbb{P}_2$  such that  $[q]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$  for the basis  $\mathcal{B} = \{p_1, p_2, p_3\}$  of  $\mathbb{P}_2$ , we need only use the change-of-coordinates matrix  $P_{\mathcal{B}} = \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix}$  in the following manner:

$$\begin{aligned} q(t) &= P_{\mathcal{B}}[q]_{\mathcal{B}} \\ &= \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \\ &= -p_1(t) + p_2(t) + 2p_3(t) \\ &= -(1 + t^2) + (t - 3t^2) + 2(1 + t - 3t^2) \\ &= -1 - t^2 + t - 3t^2 + 2 + 2t - 6t^2 \\ &= 1 + 3t - 10t^2. \end{aligned}$$

Alternatively, if you feel more comfortable working exclusively with vectors and matrices, we could also convert the basis  $\mathcal{B}$  into vectors in the standard basis of  $\mathbb{P}_2$  and use this to construct our matrix. That is, in the standard basis  $E = \{1, t, t^2\}$ , we have

$$[p_1]_E = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad [p_2]_E = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}, \quad [p_3]_E = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}.$$

So, form the matrix

$$\begin{bmatrix} [p_1]_E & [p_2]_E & [p_3]_E \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix}$$

and compute

$$\begin{bmatrix} [p_1]_E & [p_2]_E & [p_3]_E \end{bmatrix} [q]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -10 \end{bmatrix}.$$

Converting back to standard coordinates, it is easy to see that this gives the same answer as our original calculation. That is, writing  $[\mathcal{B}]_E$  to represent  $\begin{bmatrix} [p_1]_E & [p_2]_E & [p_3]_E \end{bmatrix}$ , we have the nice little formula

$$[\mathcal{B}]_E [q]_{\mathcal{B}} = [q]_E.$$

In fact, this generalizes to arbitrary sets of bases in vector spaces. Namely, if  $\mathcal{B}$  and  $\mathcal{D}$  are any two bases for some arbitrary finite dimensional vector space  $V$ , this yields the nice-looking little formula which holds for every  $x \in V$ :

$$[\mathcal{B}]_{\mathcal{D}} [x]_{\mathcal{B}} = [x]_{\mathcal{D}}.$$